

DRY-RUN GUIDE | 19_floyd_cycle_detection.py
Source: Python\dsa_solutions\19_floyd_cycle_detection.py

CORE BEHAVIOR

Uses slow (one link) and fast (two links) pointers; a meeting indicates a cycle.

REPRESENTATIVE INPUT

1->2->3->2 (loop to node 2).

STEP TRACE

Pointers advance; on a cycle their positions eventually coincide.

EXPECTED OUTPUT / RETURN

Cycle detected / source-specific Boolean.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\dsa_solutions\19_floyd_cycle_detection.py; line numbers refer to the original file. No source was modified.

```
001 | import sys
002 |
003 |
004 | class Node:
005 |     def __init__(self, value):
006 |         self.value = value
007 |         self.next = None
008 |
009 |
010 | def has_cycle(head):
011 |     slow = fast = head
012 |     while fast and fast.next:
013 |         slow = slow.next
014 |         fast = fast.next.next
015 |         if slow is fast:
016 |             return True
017 |     return False
018 |
019 |
020 | def main():
021 |     data = list(map(int, sys.stdin.read().split()))
022 |     if not data:
023 |         return
024 |     count = data[0]
025 |     values = data[1:1 + count]
026 |     position = data[1 + count]
027 |     nodes = [Node(value) for value in values]
028 |     for index in range(count - 1):
029 |         nodes[index].next = nodes[index + 1]
030 |     if count and position != -1:
031 |         nodes[-1].next = nodes[position - 1]
032 |     print("true" if has_cycle(nodes[0] if nodes else None) else "false")
033 |
034 |
035 | if __name__ == "__main__":
036 |     main()
037 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.